Myth or fact? Energy quick-check

Year 8 SEAL Science · Chapter 7 Energy · 20 statements, about 15 minutes

Name: _____ Class: _____ Date: _____

What to do. Read each statement. Circle FACT if it is correct, or MYTH if it is wrong. If it is a myth, write the correct idea on the line. Mark your own answers using the last page.

1	Energy is a substance, like a liquid you could pour into a jar. Lesson 1	**FACT / MYTH**
If it is a myth, the correct idea: _____		

2	A squashed spring has no energy because it is not moving. Lesson 2	**FACT / MYTH**
If it is a myth, the correct idea: _____		

3	If a remote-control car goes three times as fast, its kinetic energy becomes three times bigger. Lesson 2	**FACT / MYTH**
If it is a myth, the correct idea: _____		

4	A heavier object always has more kinetic energy than a lighter one. Lesson 2	**FACT / MYTH**
If it is a myth, the correct idea: _____		

5	A 2 kg pot plant on a balcony shelf 1.5 m up has energy, even though it is not moving. Lesson 2	**FACT / MYTH**
If it is a myth, the correct idea: _____		

6	When a hot object and a cold object touch, heat flows from the cold one into the hot one. Lesson 3	**FACT / MYTH**
If it is a myth, the correct idea: _____		

7	Convection can happen inside a solid block of metal. Lesson 3	**FACT / MYTH**
If it is a myth, the correct idea: _____		

8	Heat from the Sun reaches Earth by radiation, even though space is almost empty. Lesson 3	**FACT / MYTH**

If it is a myth, the correct idea:

9	A good insulator stops heat moving completely. Lesson 3	**FACT / MYTH**

If it is a myth, the correct idea:

10	When a toy car rolls across the carpet and stops, its energy has been destroyed. Lesson 4	**FACT / MYTH**

If it is a myth, the correct idea:

11	An energy transfer means energy changes from one type into another. Lesson 4	**FACT / MYTH**

If it is a myth, the correct idea:

12	A desk fan takes in 100 J of electrical energy and gives out 70 J of kinetic energy. It is 70% efficient. Lesson 4	**FACT / MYTH**

If it is a myth, the correct idea:

13	In a Sankey diagram, the output arrows add up to the same width as the input arrow. Lesson 4	**FACT / MYTH**

If it is a myth, the correct idea:

14	When you eat a muesli bar, your body makes new energy out of nothing. Lesson 4	**FACT / MYTH**

If it is a myth, the correct idea:

15	First Nations Peoples have only used fire for a few hundred years. Lesson 5	**FACT / MYTH**

If it is a myth, the correct idea:

16	Watts tell you how much energy an appliance has used altogether. Lesson 6	**FACT / MYTH**

If it is a myth, the correct idea:

17	A 2,000 W hair dryer used for 1 hour uses the same energy as a 1,000 W iron used for 2 hours. Lesson 6	**FACT / MYTH**
If it is a myth, the correct idea: _____		

18	Insulation with a higher R-value slows heat flow more than insulation with a lower R-value. Lesson 7	**FACT / MYTH**
If it is a myth, the correct idea: _____		

19	'Reliable' and 'valid' mean the same thing in an investigation. Lesson 8	**FACT / MYTH**
If it is a myth, the correct idea: _____		

20	A system has enough energy when the supply is equal to or bigger than the demand. Lesson 9	**FACT / MYTH**
If it is a myth, the correct idea: _____		

My score: _____ / 20. Which lessons did your wrong answers come from? Revise those first with the "I can" checklist.

Answers: mark your own

Give yourself 1 mark for the right choice. For a myth, give yourself a bonus tick if your correct idea matches the reason below.

#	Answer	Why
1	**MYTH**	Energy is not a substance. It is the ability to make things happen or change. Fuels and batteries store energy, but energy itself is not stuff.
2	**MYTH**	A squashed spring stores elastic potential energy. Let it go and that energy becomes movement.
3	**MYTH**	Speed is squared in KE = ½mv². Three times the speed gives $3^2 = 9$ times the kinetic energy.
4	**MYTH**	Kinetic energy depends on mass and speed. A 0.06 kg ball at 50 m/s has 75 J. A 60 kg person walking at 1 m/s has only 30 J.
5	**FACT**	It has gravitational potential energy because of its height. GPE = mgh = 2 × 10 × 1.5 = 30 J.
6	**MYTH**	Heat flows from the hotter object to the colder one. A cold drink in a warm room warms up because heat flows from the room into the drink.
7	**MYTH**	Convection needs a fluid, which is a liquid or a gas, so the particles can flow. In a solid, heat moves by conduction.
8	**FACT**	Radiation travels as waves and needs no particles. Conduction and convection both need particles, so they can't cross empty space.
9	**MYTH**	Insulators only slow heat down. Given enough time, heat still gets through, so a frozen pie in a cooler bag still thaws in the end.
10	**MYTH**	Energy is never destroyed. The car's energy is still there. It has spread out into the room as a tiny bit of warming.
11	**MYTH**	That describes a transformation. A transfer moves energy from one object to another, without changing its type.
12	**FACT**	70 ÷ 100 × 100% = 70%. The other 30 J is wasted, mostly as thermal energy and sound.
13	**FACT**	Energy is conserved, so all the outputs together must equal the input. Arrow widths show the amounts.
14	**MYTH**	Energy can't be made from nothing. The energy was already stored in the food, and your body transforms it.
15	**MYTH**	First Nations Peoples have used fire for tens of thousands of years, and still do today.
16	**MYTH**	Watts measure power, which is how fast energy is used. The energy used altogether is in Wh or kWh, so you also need the time.
17	**FACT**	Hair dryer: 2,000 W × 1 h = 2,000 Wh. Iron: 1,000 W × 2 h = 2,000 Wh. Both use 2 kWh.
18	**FACT**	A higher R-value means better insulation, so heat passes through it more slowly.
19	**MYTH**	Reliable results come out about the same when the test is repeated. A valid test is a fair test that measures what it claims to.
20	**FACT**	If the demand is bigger than the supply, the system runs short of energy.